Students as Curriculum Architects, AI as Scaffolding Partner

EL Education 2026 National Conference: Session Details

Presenters: Aaron Pomeranz, PsyD (School Psychologist, King Middle School / Portland Public Schools); Lisa Hatch (7th Grade ELA, King Middle School); Brian Nail (Special Education / Resource Room, King Middle School)
Format: 90-minute hands-on workshop
Document contents: Learning Targets and Session Outline with embedded timing

Learning Targets

Participants will leave this session able to take three concrete actions.

1. I can articulate the research basis for student-led teaching (the Protégé Effect, UDL, and peer-mediated instruction), and explain how AI scaffolding amplifies these benefits at exactly the points where AI cannot do the work alone.

Anchor segment: The Research (minutes 5 to 15)
Anchor citations: Chase et al. (2009); Fiorella & Mayer (2013); CAST UDL Guidelines (2024)
Take-home: language to use with administrators, parents, and colleagues for distinguishing AI's content-generation role from the audience-modeling work only students can do.

2. I can use AlloFlow to transform a student-authored (or sample-authored) source text into a multi-modal, UDL-aligned lesson, apply EL's critique protocols to a partner's transformation choices, verify source-grounding and run a back-translation check on the multilingual version (with the consultative classroom protocol named explicitly), and experience the live-polling and auto-routing tools peer-authors use to differentiate instruction in real time.

Anchor segment: Hands-On Build (minutes 25 to 50)
Take-home: a critiqued, verified, production-ready multi-modal artifact on the participant's device, plus first-hand experience of the live-polling auto-routing mechanism that drives in-the-moment differentiation during the TEACH phase of the cycle.

> **3. I can design a "Students as Curriculum Architects" unit for my own classroom or school** in which each student authors two lessons (one academic and one peer-shareable SEL skill lesson); specify audience, assessment, access supports, and the teacher's role at each step of the six-step *author, transform, critique, verify, curate, teach* cycle (including the live-delivery routing and polling decisions); and identify my first three implementation steps.
>
> **Anchor segments:** Design Your Own Unit (minutes 65 to 80) and Contextualize & Next Steps (minutes 80 to 87)
>
> **Take-home:** a populated planning template, a written next-step commitment posted publicly during the gallery share, and connection to the AlloFlow open-source community for ongoing peer-mentor support.

Session Outline

Time Overview

Time	Segment
0 to 5 min	Welcome & Hook
5 to 15 min	The Research
15 to 25 min	The Model & The Pilot
25 to 50 min	Hands-On Build
50 to 65 min	Share-Out & Content Domain Discussion
65 to 80 min	Design Your Own Unit
80 to 87 min	Contextualize & Next Steps
87 to 90 min	Closing

0 to 5 min · Welcome & Hook

We introduce ourselves and the workshop's central question: **What happens when students become curriculum architects?** A quick live poll (show of hands or shared digital response) asks participants how often their own students currently produce instructional content for peers. The data sets the room and motivates the rest of the session.

5 to 15 min · The Research

Through slide deck and live polling, we introduce three research strands:

- **Protégé Effect** (Chase et al., 2009; Fiorella & Mayer, 2013): preparing to teach drives deeper learning than studying for assessment.

- **Universal Design for Learning** (CAST UDL Guidelines, 2024): designing for variability is the foundation of access.
- **Peer-mediated instruction and intervention**: peer-tutor / peer-learner configurations produce gains across cognitive, social, and behavioral measures.

The segment closes by introducing the proposal's central insight: **AI's pedagogical value is greatest at exactly the points where AI cannot do the work alone.** AI produces competent-looking baseline material in any domain, but it cannot replace the student's authorship work, which in both academic and SEL lessons combines source-grounded factual verification with audience-aware register, examples, and framing for the actual peers in the room.

15 to 25 min · The Model & The Pilot

Walkthrough of the six-step cycle students follow in the model:

AUTHOR → TRANSFORM → CRITIQUE → VERIFY → CURATE → TEACH

Step	Students	Teacher	AI	School Psychologist	Special Educator
AUTHOR	Locate and evaluate own sources. Submit source list to teacher for credibility review. Draft source text grounded in that reading and (for SEL) peer-audience knowledge.	Selects unit, content domain, audience, quality bar. Reviews each student's source list for credibility and approves the student to draft.	Inactive at this stage. Optional diagnostic (Analyze Source Material) once a draft exists, surfacing reading level, key concepts, and factual accuracy with citations.	For SEL units, defines the scope of teachable peer skills and excludes clinical or crisis topics from the audience-facing curriculum.	Coaches own students as peer-authors, using the graduated-support spectrum (Author-first / AI-revision-support / full generation) to scaffold authorship at each student's writing-readiness level. Also advises gen-ed peer-authors on prerequisites, scaffolding, and audience-fit for classmates with IEPs.
TRANSFORM	Run authored text through AlloFlow's multi-modal pipeline (audio, leveled, multilingual, visual). Select accessibility supports for known audience needs.	Decides which students need AI revision support; reviews any AI-refined source text before downstream resources generate.	Generates audio, leveled differentiated content, multilingual versions, and visual organizers from the human-authored source. Optionally refines source text in response to a student's written revision instruction.	Not active unless flagged by the teacher.	Reviews own students' transformation choices alongside the peer-author. Also reviews access supports in gen-ed lessons against IEP goals; flags additional scaffolding the peer-author may not see firsthand.

Step	Students	Teacher	AI	School Psychologist	Special Educator
CRITIQUE	Apply "I notice / I wonder" and "Be Kind / Be Specific / Be Helpful" protocols to partner work.	Designs or adapts the rubric; facilitates protocols.	Inactive. Peer critique is human work the platform cannot do.	Not active.	Facilitates critique within the resource-room cohort; supports cross-setting critique pairs when scheduling allows.
VERIFY	Source-ground any AI-augmented elements. For leveled outputs, check meaning preservation. For multilingual outputs, consult a proficient reader (multilingual classmate, family member, ELL teacher, district multilingual staff) before the version reaches peers; where no proficient reader is available, run a back-translation check and flag for staff review.	Models source evaluation, source-grounding verification, and the consultative translation-fidelity protocol before students attempt them.	Subject of inspection, not actor: produced the outputs being verified.	Not active.	Models verification with own student-authors. Also reviews access supports in gen-ed lessons against IEP goals; confirms differentiated outputs meet planned scaffolding needs.
CURATE	Submit finished lesson for teacher review; receive approval and feedback before publication; participate in Models of Excellence selection.	Reviews lessons before they reach audience; gives final approval. Selects Models of Excellence.	Inactive. Curatorial judgment is human work.	For SEL content, conducts clinical review and gives final go or no-go before any lesson reaches a peer audience. Not active for non-SEL academic units.	Curates own students' lessons for publication; consults on gen-ed lessons when the planned audience includes students whose IEP goals intersect with the content.
TEACH	Lead the lesson live for the peer audience; use polling to read the room in real time; auto-route consumer-peers into themed or scaffolded groups based on their responses; respond to questions and adjust pacing.	Coaches the peer-author through differentiation choices before delivery; observes the live teaching moment without intervening; debriefs with the peer-author after.	Provides live-session infrastructure (live polling, auto-routing rules, per-group resource delivery). Response data flows directly between student devices and is never persisted on third-party servers. Does not lead the lesson.	For SEL content, present in the room and available to the peer-author if a consumer-peer raises a safety-boundary topic. Otherwise passive.	Coaches own peer-authors through live delivery, often to familiar or smaller audiences as a starting point. Also present at gen-ed delivery when the audience includes a student with an IEP goal directly addressed by the lesson.

Brief overview of the King Middle pilot context. The pilot spans Lisa's 7th grade ELA section, which includes multilingual learners and classmates from specialized special education settings who join expeditions alongside their grade-mates, and Brian's resource-room cohort, whose students author lessons in the same unit. AlloFlow's graduated-support spectrum (Author-first, Author-with-AI-revision-support, and full generation as a UDL writing accommodation) lets every student enter authorship at the scaffolding level that works for them. Across the unit, each student authors two lessons (one academic and one peer-shareable SEL skill lesson). School-psychologist clinical oversight applies to the SEL content domain. The special educator on the team coaches the resource-room peer-authors directly and also advises gen-ed peer-authors on access supports, IEP-goal alignment, and the bridge between general-ed lessons and the targeted instruction his students continue to receive in the resource room.

25 to 50 min · Hands-On Build (Path 3 structure)

The 25-minute hands-on segment is paced through four tight phases:

- **~5 min, Brief generator demonstration.** Participants prompt AlloFlow with a topic and observe what the platform generates from scratch. Explicitly framed: "*This is what the tool can do as a content generator. In our pilot at King, this is NOT how students typically start, but it's useful for you to see what AlloFlow is capable of.*"
- **~10 min, Author-first practice (the pilot's actual workflow).** Participants take either a pre-written instructional text they brought from their own curriculum or one of three sample source texts provided at varying grade bands. They run their authored text through AlloFlow's transformation pipeline, generating audio narration, leveled differentiated content, multilingual versions, and visual organizers from the human-authored source. Participants who want to see the graduated-support spectrum can optionally invoke the platform's Analyze Source Material diagnostic and the AI-revision-support path (a third option for students who need scaffolded revision before downstream generation); the teacher gates which students use this path in the pilot.
- **~5 min, Critique.** Participants apply EL's "I notice / I wonder" protocol to a partner's transformation choices, focusing on accessibility decisions and audience fit.
- **~5 min, Verify.** Participants check source-grounding on one AI-augmented element (do the platform's surfaced citations actually support the claim?) and run a back-translation check on the multilingual version. The presenters explicitly name what this exercise stands in for: in the classroom, the multilingual fidelity check is consultative work between the student-author and a proficient reader of the target language (multilingual classmate, family member, ELL teacher, district multilingual staff), not something the author does alone. The workshop's back-translation step is a partial substitute participants can use immediately when no proficient reader is available, and it doubles as a model for how to flag a translation for staff review.

All three presenters and table partners circulate throughout the room, supporting peer critique at each table. No prior AlloFlow experience is required. By minute 50, every participant has a transformed, critiqued, verified, production-ready artifact on their device. During this segment we also demonstrate the live-polling tool: participants receive a short in-the-moment poll using AlloFlow's live-session connections, experiencing the same auto-routing-by-response mechanism that peer-authors use to differentiate instruction live. The response data flows directly between participant devices and is never persisted on third-party servers. The demo doubles as a metacognitive layer (participants experience what students experience) and a working proof point.

50 to 65 min · Share-Out & Content Domain Discussion

Open discussion structured in three short rounds:

- **Round 1, Show & Tell (~5 min):** three or four participants briefly share what their transformed artifact looks like, focusing on accessibility choices they made.
- **Round 2, Content Domain Parity (~5 min):** we surface the model's central framing that both academic content and peer-shareable SEL skill content require the same authoring work (source-grounded factual verification plus audience-aware register, examples, and framing). Participants discuss in pairs how the two domains would play out in their own current curriculum, and what the comparison between the two finished lessons might reveal about each student's mastery.
- **Round 3, Safety Boundary (~5 min):** brief overview of the clinical boundary for SEL content. Teachable peer skills (active listening, conflict mediation, asking for help, growth mindset framing, naming emotions) are within scope; clinical or crisis content (suicidal ideation, abuse disclosure, trauma processing) is explicitly routed to clinical staff. The boundary is curated by the school psychologist on the pilot team; the planning template includes guidance for adopters without school-psychologist staffing.

65 to 80 min · Design Your Own Unit

Planning template handed out. Each participant works individually on the same structured fields students use:

- Audience (within-class peers / cross-classroom / cross-age / families / community)
- Lesson pair (one academic + one peer-shareable SEL skill lesson per student) and standards alignment for each
- Source-evaluation protocol (how students locate and vet sources; how the teacher reviews source lists for credibility before drafting)
- Access supports to scaffold (multilingual / leveled / audio / visual / SEL-specific)
- Verification protocol (source-grounding check; consultative translation-fidelity check with proficient reader; clinical curation if SEL)

- Inclusive and co-teaching context (general-ed and resource-room coordination; IEP-goal alignment; access-need anticipation)
- Teacher role at each of the six cycle steps (including the TEACH-phase polling and auto-routing decisions)
- First three implementation steps for the next school week

Table-group share creates accountability for completing each field. Participants leave with a draft unit plan, not just an idea.

80 to 87 min · Contextualize & Next Steps

Gallery share: each participant posts their written next-step commitment publicly on chart paper or shared digital wall. Resource sharing:

- AlloFlow GitHub repository and Gemini Canvas link
- AlloFlow open-source community channels
- The King pilot team (school psychology, ELA, and special education) available as a peer-mentor resource during the first year for participants implementing the model in their own contexts

87 to 90 min · Closing

We restate the three Learning Targets, name one moment from the session that captured each, thank participants, and invite ongoing dialogue. We remain in the room afterward for questions and individual planning support.

Notes for facilitators: Path 3 timing (the four hands-on phases at 5/10/5/5 minutes) is tight; the generator demonstration phase can compress to 3 minutes if needed. The 50 to 65 min Share-Out segment is flexible. Round 3 (Safety Boundary) can extend if SEL content interest is high, or compress if the room is mostly focused on academic content.